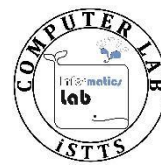




Institut Sains dan Teknologi Terpadu Surabaya

Jl. Ngagel Jaya Tengah 73 - 77, Surabaya 60284

Telp. (031) 5027920 Fax. (031) 5041509



Laboratorium : E-401

Waktu : Rabu / 13.15-15.15

Minggu Ke : 4

Materi : Inheritance

Praktikum : Pemrograman Berorientasi Objek

Jurusan : S1 - Informatika

Tanggal : 1 April 2026

Jenis Soal : Materi dan Tugas

MATERI (TOTAL: 35)

Buatlah game pokemon simple. Saat program pertama kali dijalankan, minta player memilih starternya.

```
=== Pocket Monster ===
Choose your starter
1. Charmander
2. Squirtle
3. Bulbasaur
>> 
```

Tabel berikut menunjukkan Pokemon yang dapat dipilih:

Nama Pokemon	HP	Attack	Tipe
Charmander	100	10	Fire (Api)
Squirtle	100	10	Water (Air)
Bulbasaur	100	10	Grass (Rumput)

Setelah memilih starter, program akan secara otomatis menentukan pokemon milik lawan (rival) secara acak, dengan ketentuan pokemon lawan tidak boleh sama dengan pilihan player. Tampilkan pesan pilihan pokemon baik milik player maupun lawan. Lalu, game akan masuk ke mode battle dengan menampilkan informasi pokemon seperti nama, hp, attack dan tipe pokemon.

```
>> 1
You choose Charmander
Your Rival choose Bulbasaur

=====
| Player                | Rival                |
=====
| Name: Charmander      | Name: Bulbasaur      |
| HP: 100/100           | HP: 100/100          |
| Attack: 10            | Attack: 10           |
| Type: Fire            | Type: Grass           |
=====

1. Attack
2. Run
Choose your option: 
```

Player memiliki dua opsi, yaitu attack atau run. Jika player memilih run akan langsung menghentikan program. Jika player memilih attack, pokemon milih player akan menyerang lawan. Damage yang diberikan sebesar attack dari pokemon yang menyerang dikali multiplier (attack * multiplier). Setelah itu, pokemon lawan otomatis melakukan serangan balik kepada player dengan mekanisme yang sama.

Nilai multiplier ditentukan berdasarkan kecocokan tipe Pokémon penyerang dan lawan. Multiplier ini mempengaruhi besar kecilnya damage yang diberikan. Klasifikasi multiplier sebagai berikut:

Pesan Efektivitas	Multiplier	Rumus Damage
Super effective	2	attack * 2
Not very effective	0.5	attack * 0.5

Misalnya: Jika Squirtle (Water) menyerang Charmander (Fire):

- Tipe Water unggul terhadap Fire
- Maka multiplier = 2
- Damage yang diberikan menjadi dua kali lipat
- Program menampilkan pesan: It's super effective!

Berikut adalah contoh pada program, setiap serangan menampilkan nama skill, efektivitas, jumlah damage yang diterima oleh lawan.

```
Choose your option: 1

--- You attack ---
Charmander use Ember
It's super effective!
Bulbasaur takes 20 damage!

--- Rival attacks ---
Bulbasaur use Razor Leaf
It's not very effective...
Charmander takes 5 damage!
```

Berikut adalah nama skill untuk masing-masing pokemon:

Pokemon	Nama Skill
Charmander	Ember
Squirtle	Water Gun
Bulbasaur	Razor Leaf

Berikut adalah tabel matchup tipe.

Attacker	Defender	Efek	Multiplier
Fire	Grass	Super effective	×2
Fire	Water	Not effective	×0.5
Water	Fire	Super effective	×2
Water	Grass	Not effective	×0.5
Grass	Water	Super effective	×2
Grass	Fire	Not effective	×0.5

Intinya:

- Fire (Api) mengalahkan Grass (Rumput)
- Grass (Rumput) mengalahkan Water (Air)
- Water (Air) mengalahkan Fire (Api)

Jika HP lawan ≤ 0 , maka player menang. Sedangkan jika HP player ≤ 0 , maka player kalah. Setelah player menang atau kalah, hentikan program.

Hint: Pada subclass, misal: Bulbasaur.java, ada function attackCharmander dan attackSquirtle.

**DILARANG MENGGUNAKAN MATERI YANG BELUM DIAJARKAN PADA MINGGU INI
JIKA MELANGGAR MAKA NILAI MATERI : 0**

PERHATIKAN KETENTUAN DIBAWAH :

- Highlight kriteria yang dikerjakan dengan warna kuning dan kumpulkan word beserta dengan file materi, apabila tidak dikumpulkan maka materi tidak akan diperiksa.
- Akan ada pengurangan nilai sebesar -5 untuk setiap kriteria yang dihighlight namun tidak dikerjakan.
- **MENCONTEK = Nilai MOD 2**

MATERI : 35 (+5)

SCORE	KRITERIA
0/2	Dapat memilih starter pokemon
0/1/3	Dapat mengacak pokemon milik musuh @1 kalau bisa sama dengan pokemon player
0/2	Menampilkan informasi pokemon player
0/2	Menampilkan informasi pokemon lawan
0/2	HP pokemon berkurang setelah attack
0-3	Nama skill pokemon sesuai @1 per pokemon
0-6	Semua kasus (jumlah damage) super effective benar (Charmaner, Squirtle, Bulbasaur) @2 per pokemon
0-6	Semua kasus (jumlah damage) not very effective benar (Charmander, Squirtle, Bulbasaur) @2 per pokemon
0/3	Lawan menyerang secara otomatis.
0/2	Dapat melakukan aksi run (Program berhenti)
0/2	Terdapat pengecekan player menang dan kalah
0/2	Program berhenti setelah player menang atau kalah.
-10/0	Menggunakan konsep inhertance (-10 kalau tidak menggunakan inheritance)
-35/+5	Tidak menggunakan konsep polymorphism (-35 kalau menggunakan konsep polymorphism)
Total : 35 (+5)	

TUGAS (TOTAL = 35)

Buatlah game Clash Royale

Menu Awal

Saat program pertama kali dijalankan, tampilkan judul dan pilihan untuk bermain (Play) dan menghentikan program (Exit).

```
=== Clash Royale ===
1. Play
2. Exit
>> █
```

Menu Game

Pada menu game, tampilkan menu seperti gambar dibawah, terdapat arena, informasi elixir player dan opsi pilihan untuk player. Di awal permainan, player dan enemy masing-masing memiliki 3 tower: 1 King Tower (tengah) yang memiliki HP 20, 2 Princess Tower (kiri dan kanan) yang masing-masing memiliki HP 10. Player memulai permainan dengan 5 elixir.

```
=====
|          CLASH ROYALE          |
=====
|          [Enemy]              |
| [PT]   [KT]   [PT]           |
| [10]   [20]   [10]           |
|----- RIVER -----|
| [10]   [20]   [10]           |
| [PT]   [KT]   [PT]           |
|          [Player]             |
=====
Elixir: 5/10
Choose a card:
1. Goblins [COST: 2]
2. Goblin Hut [COST: 5]
3. Fireball [COST: 4]
4. Skip Turn
>> █
```

Program akan merandom 3 kartu untuk player mainkan: 1 kartu Troop, 1 kartu Building dan 1 kartu Spell. Masing-masing menampilkan Elixir Costnya mereka. Player dapat memilih salah satu dari ketiga kartu random ini atau skip turn. Memilih kartu akan mengurangi elixir player sesuai dengan Elixir Cost setiap kartu, berikan pengecekan elixir player cukup untuk memainkan kartu yang dipilih. Setelah memilih kartu, pada turn berikutnya, elixir player akan bertambah 1. Berikut adalah tabel dari semua kartu yang ada:

Jenis Kartu	Nama Kartu	HP	Damage	Elixir Cost	Deskripsi
Troop	Goblins	3	1	2	Saat dipilih, spawn 2–5 goblin. Setiap goblin menyerang target (Troop > Building > Princess Tower > King Tower) dengan damage 1.
	Elixir Golem	5	2	3	Saat mati, pecah menjadi 2 Mini Elixir Golem (HP: 2, Damage: 1). Saat Mini Golem mati, player mendapat +1 elixir.
	Bomber	4	3	4	Menyerang target dengan damage 3. Saat mati, memberikan damage

Jenis Kartu	Nama Kartu	HP	Damage	Elixir Cost	Deskripsi
					ledakan sebesar 2 ke semua troop, bangunan dan tower lawan.
Building	Cannon	5	-	4	Menyerang tower musuh setiap turn (2 damage)
	Elixir Collector	5	-	4	Memberikan +1 elixir setiap turn
	Goblin Hut	5	-	5	Spawn 2–4 goblin setiap turn
Spell	Fireball	-	-	4	Memberi 4 damage ke 1 tower yang dipilih
	Arrow	-	-	3	Memberi 2 damage ke semua tower musuh
	Goblin Barrel	-	-	3	Memberi 2 damage pada tower yang dipilih. dan akan spawn 1–3 goblin yang menyerang tower yang dipilih.

Kartu Troop

Semua kartu troop akan menyerang target dengan urutan prioritas. Ketika terdapat troop musuh di arena, troop akan menyerang troop tersebut terlebih dahulu hingga salah satu unit mati. Jika tidak terdapat troop musuh, maka troop akan menyerang Building musuh. Apabila terdapat lebih dari satu Building, target akan dipilih secara acak. Jika tidak terdapat troop maupun Building, maka troop akan menyerang Tower, dimulai dari Princess Tower secara acak, dan jika kedua Princess Tower telah hancur, maka akan menyerang King Tower. Pada menu game, akan muncul troops yang masih hidup di bagian active troops (Player dibawah, Enemy diatas). Semua troop yang masih hidup akan tetap terus menyerang musuh setiap turn sampai mati.

Tabel berikut menunjukkan contoh ketika menggunakan kartu troop

Kartu	Contoh di Program	Penjelasan
Goblins	Contoh 1: <pre> ===== TURN 1 ===== >> Player deploys Goblins(4) >> Enemy skipped their turn. ----- [ATTACK] Player Goblin => Enemy Right Princess Tower (1 dmg) [COUNTER] Enemy Right Princess Tower => Goblin (3 dmg) [DEATH] Player Goblin died from counter attack! [ATTACK] Player Goblin => Enemy Left Princess Tower (1 dmg) [COUNTER] Enemy Left Princess Tower => Goblin (3 dmg) [DEATH] Player Goblin died from counter attack! [ATTACK] Player Goblin => Enemy Left Princess Tower (1 dmg) [COUNTER] Enemy Left Princess Tower => Goblin (3 dmg) [DEATH] Player Goblin died from counter attack! [ATTACK] Player Goblin => Enemy Left Princess Tower (1 dmg) [COUNTER] Enemy Left Princess Tower => Goblin (3 dmg) [DEATH] Player Goblin died from counter attack! ===== ===== CLASH ROYALE ===== [Enemy] [PT] [KT] [PT] [07] [20] [09] ----- RIVER ----- [10] [20] [10] [PT] [KT] [PT] [Player] ===== </pre>	Contoh 1: • Player memilih goblin • Enemy skip turn • [ATTACK] Goblin milik player menyerang salah satu princess tower (secara acak), dengan damage sebesar 1 • [COUNTER] Tower milik enemy menyerang kembali, dengan damage sebesar 3 • [DEATH] Goblin milik player mati (karena HPnya hanya 3) • Karena ada 3 goblin, pesannya diulang 3 kali. Contoh 2: • Enemy memiliki Elixir golem yang masih aktif/hidup • Player memiliki Cannon yang masih aktif/hidup • Player deploy goblin • Enemy deploy fireball

Kartu	Contoh di Program	Penjelasan
	Contoh 2: <pre> ===== CLASH ROYALE ===== == Enemy Active Troops == - Elixir Golem (HP: 5) ===== [Enemy] [PT] [KT] [PT] [10] [20] [08] ----- RIVER ----- [07] [20] [09] [PT] [KT] [PT] [Player] ===== == Player Active Buildings == - Cannon (HP: 3) ===== Elixir: 3/10 Choose a card: 1. Goblins [COST: 2] 2. Cannon [COST: 4] 3. Goblin Barrel [COST: 3] 4. Skip Turn >> 1 ===== TURN 3 ===== >> Player deploys Goblins(4) >> Enemy deploys Fireball [EFFECT] Fireball deals 4 damage to Player King Tower ----- [Player Building] Cannon attacks Enemy Right Princess Tower(2 dmg) ----- [ATTACK] Player Goblin => Enemy Elixir Golem (1 dmg) [ATTACK] Player Goblin => Enemy Elixir Golem (1 dmg) [ATTACK] Player Goblin => Enemy Elixir Golem (1 dmg) [ATTACK] Player Goblin => Enemy Elixir Golem (1 dmg) [ATTACK] Enemy Elixir Golem => Player Goblin (2 dmg) ===== </pre>	• Fireball dari enemy menyerang king tower player • Cannon milik player menyerang tower milik enemy • Goblin milik player menyerang elixir golem milik enemy • Elixir golem milik enemy menyerang goblin player.
Elixir Golem	<pre> ===== TURN 1 ===== >> Player deploys Elixir Golem >> Enemy skipped their turn. ----- [ATTACK] Player Elixir Golem => Enemy Right Princess Tower (2 dmg) [COUNTER] Enemy Right Princess Tower => Elixir Golem (3 dmg) ===== CLASH ROYALE ===== [Enemy] [PT] [KT] [PT] [10] [20] [08] ----- RIVER ----- [10] [20] [10] [PT] [KT] [PT] [Player] ===== == Player Active Troops == - Elixir Golem (HP: 2) ===== </pre>	• Player memilih Elixir Golem • Enemy skip turn • [ATTACK] Elixir golem menyerang salah satu princess tower (secara acak), dengan damage sebesar 2 • [COUNTER] Tower menyerang kembali dengan damage sebesar 3. • Elixir golem hidup dengan HP 2 Turn berikutnya: • Player skip turn • Enemy skip turn • [ATTACK] Elixir golem menyerang tower (secara acak) • [COUNTER] Tower menyerang kembali • [DEATH] Elixir golem mati

Kartu	Contoh di Program	Penjelasan
	<pre> ===== TURN 2 ===== >> You skipped Player turn. >> Enemy skipped their turn. ----- [ATTACK] Player Elixir Golem => Enemy Left Princess Tower (2 dmg) [COUNTER] Enemy Left Princess Tower => Elixir Golem (3 dmg) [DEATH] Player Elixir Golem died from counter attack! [EFFECT] Elixir Golem split into 2 Mini Golems! [ATTACK] Player Mini Elixir Golem => Enemy Left Princess Tower (1 dmg) [COUNTER] Enemy Left Princess Tower => Mini Elixir Golem (3 dmg) [DEATH] Player Mini Elixir Golem died from counter attack! [EFFECT] Mini Golem died => Player gains +1 Elixir [ATTACK] Player Mini Elixir Golem => Enemy Right Princess Tower (1 dmg) [COUNTER] Enemy Right Princess Tower => Mini Elixir Golem (3 dmg) [DEATH] Player Mini Elixir Golem died from counter attack! [EFFECT] Mini Golem died => Player gains +1 Elixir ===== CLASH ROYALE ===== [Enemy] [PT] [KT] [PT] [07] [20] [07] ----- RIVER ----- [10] [20] [10] [PT] [KT] [PT] [Player] ===== Elixir: 6/10 </pre>	• [EFFECT] Elixir golem pecah jadi 2 mini elixir golem • [ATTACK] Mini elixir golem menyerang tower • [COUNTER] tower menyerang kembali • [DEATH] mini elixir golem mati • [EFFECT] mini elixir golem memberikan 1 elixir. • Karena ada 2 mini elixir golem, pesannya diulang 2 kali. • Elixir player bertambah 2 (dari 2 mini elixir golem)
Bomber	<pre> ===== TURN 4 ===== >> Player deploys Bomber >> Enemy skipped their turn. ----- [ATTACK] Player Bomber => Enemy Left Princess Tower (3 dmg) [COUNTER] Enemy Left Princess Tower => Bomber (3 dmg) ===== CLASH ROYALE ===== [Enemy] [PT] [KT] [PT] [07] [20] [10] ----- RIVER ----- [10] [20] [10] [PT] [KT] [PT] [Player] ===== == Player Active Troops == - Bomber (HP: 1) ===== </pre>	• Player memilih Bomber • Enemy skip turn • [ATTACK] bomber menyerang tower (secara acak) dengan damage sebesar 3 • [COUNTER] tower menyerang kembali Turn berikutnya: • Player skip turn • Enemy skip turn • [ATTACK] bomber milik player menyerang tower • [COUNTER] tower milik enemy menyerang kembali • [DEATH] bomber milik player mati • [EFFECT] bomber meledaki semua troop, bangunan, dan tower lawan dengan damage sebesar 2.

Kartu	Contoh di Program	Penjelasan
	<pre> ===== TURN 5 ===== >> You skipped Player turn. >> Enemy skipped their turn. ----- [ATTACK] Player Bomber => Enemy Right Princess Tower (3 dmg) [COUNTER] Enemy Right Princess Tower => Bomber (3 dmg) [DEATH] Player Bomber died from counter attack! [EFFECT] Bomber exploded on everyone! (2 dmg) ===== CLASH ROYALE ===== [Enemy] [PT] [KT] [PT] [05] [18] [05] ----- RIVER ----- [10] [20] [10] [PT] [KT] [PT] [Player] ===== </pre>	

Kartu Building

Kartu building bersifat berbeda dari troop dan spell, building akan tetap berada di arena (aktif) selama belum hancur dan akan memberikan efek secara otomatis setiap turn (sesuai deskripsi). Pada menu game, akan muncul building yang ditempatkan di bagian active buildings (Player dibawah, Enemy diatas). Troop musuh akan memprioritaskan menyerang Building. Jika ada Building, Troop tidak langsung menyerang tower. Jika Building hancur, Troop kembali menyerang tower. Jika HP building = 0, building akan dihancurkan dan hilang dari arena.

```

=====
CLASH ROYALE
=====
= Enemy Active Buildings =
- Cannon (HP: 3)
- Elixir Collector (HP: 5)
=====
[Enemy]
[PT] [KT] [PT]
[10] [20] [10]
-----
RIVER
-----
[06] [20] [05]
[PT] [KT] [PT]
[Player]
=====
== Your Active Buildings ==
- Cannon (HP: 3)
=====
Elixir: 1/10
Choose a card:
1. Elixir Golem [COST: 3]
2. Cannon [COST: 4]
3. Fireball [COST: 4]
4. Skip Turn
>> 

```

Tabel berikut menunjukan contoh ketika menggunakan kartu Troop

Kartu	Contoh di Program	Penjelasan
Cannon	<pre> ===== TURN 1 ===== >> Player deploys Cannon >> Enemy skipped their turn. ----- [Player Building] Cannon attacks Enemy Right Princess Tower(2 dmg) ===== CLASH ROYALE ===== [Enemy] [PT] [KT] [PT] [10] [20] [08] ----- RIVER ----- [10] [20] [10] [PT] [KT] [PT] [Player] ===== == Player Active Buildings == - Cannon (HP: 5) ===== </pre>	• Player deploy cannon • Enemy skip turn • Cannon menyerang princess tower milik enemy dengan damage sebesar 2
Elixir Collector	<pre> ===== TURN 6 ===== >> Player deploys Elixir Collector >> Enemy skipped their turn. [Player Building] Elixir Collector is active! Player gained 1 Elixir. ===== CLASH ROYALE ===== [Enemy] [PT] [KT] [PT] [05] [18] [05] ----- RIVER ----- [10] [20] [10] [PT] [KT] [PT] [Player] ===== == Player Active Buildings == - Elixir Collector (HP: 5) ===== </pre>	• Player deploy elixir collector • Enemy skip turn • Elixir masih aktif, player mendapatkan 1 elixir tambahan

Goblin Hut	<pre> ===== TURN 3 ===== >> Player deploys Goblin Hut >> Enemy skipped their turn. ----- [Player Building] Goblin Hut is active! Spawning 3 Goblins! => 3 Goblins spawned! ----- [ATTACK] Player Goblin => Enemy Right Princess Tower (1 dmg) [COUNTER] Enemy Right Princess Tower => Goblin (3 dmg) [DEATH] Player Goblin died from counter attack! [ATTACK] Player Goblin => Enemy Right Princess Tower (1 dmg) [COUNTER] Enemy Right Princess Tower => Goblin (3 dmg) [DEATH] Player Goblin died from counter attack! [ATTACK] Player Goblin => Enemy Left Princess Tower (1 dmg) [COUNTER] Enemy Left Princess Tower => Goblin (3 dmg) [DEATH] Player Goblin died from counter attack! ===== ===== CLASH ROYALE ===== [Enemy] [PT] [KT] [PT] [09] [20] [08] ----- RIVER ----- [10] [20] [10] [PT] [KT] [PT] [Player] ===== == Player Active Buildings == - Goblin Hut (HP: 5) ===== </pre>	• Player deploy Goblin Hut • Enemy skip turn • Goblin Hut masih aktif, goblin hut spawn 3 goblin untuk menyerang (sama seperti kartu goblins)
------------	---	---

Kartu Spell

Kartu Spell akan langsung memberikan efek saat digunakan. Saat dipilih, player diminta untuk memilih mau menyerang ke tower mana (kecuali arrows). Tabel berikut menunjukkan contoh ketika menggunakan kartu Spell.

Kartu	Contoh di Program	Penjelasan
Fireball	<pre> ===== TURN 6 ===== Choose target tower: 1. Enemy Left Princess Tower 2. Enemy Right Princess Tower 3. Enemy King Tower >> 1 >> Player deploys Fireball [EFFECT] Fireball deals 4 damage to Enemy King Tower >> Enemy skipped their turn. ===== ===== CLASH ROYALE ===== [Enemy] [PT] [KT] [PT] [01] [18] [07] ----- RIVER ----- [10] [20] [10] [PT] [KT] [PT] [Player] ===== </pre>	• Player diminta memilih mau menyerang tower mana • Player deploy fireball • [EFFECT] fireball melakukan damage sebesar 4 pada tower yang ditarget • Enemy skip turn

Kartu	Contoh di Program	Penjelasan
Arrows	<pre> ===== TURN 1 ===== >> Player deploys Arrow [EFFECT] Arrow shower damages all target towers! >> Enemy skipped their turn. ===== CLASH ROYALE ===== [Enemy] [PT] [KT] [PT] [08] [18] [08] ----- RIVER ----- [10] [20] [10] [PT] [KT] [PT] [Player] ===== </pre>	• Player deploy arrow • [EFFECT] arrow menyerang semua tower dengan damage sebesar 2 • Enemy skip turn
Goblin Barrel	<pre> ===== TURN 2 ===== Choose target tower: 1. Enemy Left Princess Tower 2. Enemy Right Princess Tower 3. Enemy King Tower >> 1 >> Player deploys Goblin Barrel [EFFECT] Goblin Barrel hit Enemy King Tower dealing 2 damage [EFFECT] Spawning 2 Goblins on Target! >> Enemy skipped their turn. ----- [ATTACK] Player Goblin => Enemy Left Princess Tower (1 dmg) [COUNTER] Enemy Left Princess Tower => Goblin (3 dmg) [DEATH] Player Goblin died from counter attack! [ATTACK] Player Goblin => Enemy Right Princess Tower (1 dmg) [COUNTER] Enemy Right Princess Tower => Goblin (3 dmg) [DEATH] Player Goblin died from counter attack! ===== CLASH ROYALE ===== [Enemy] [PT] [KT] [PT] [05] [18] [07] ----- RIVER ----- [10] [20] [10] [PT] [KT] [PT] [Player] ===== </pre>	• Player diminta memilih mau menyerang tower mana • Player deploy Goblin Barrel • [EFFECT] Goblin Barrel melakukan damage sebesar 2 pada tower yang ditarget • [EFFECT] goblin barrel spawn 2 goblin • Enemy skip turn • Goblin menyerang enemy

Skip Turn

Jika player atau enemy memilih skip turn, player atau enemy tidak akan melakukan aksi apapun, tetapi troop atau building yang masih aktif tetap akan menyerang. Di akhir turn, player tetap akan mendapatkan 1 elixir.

Tower

Setiap kali troop menyerang tower, princess tower akan menyerang kembali dengan damage 3, sedangkan king tower akan menyerang kembali dengan damage 5.

```

===== TURN 1 =====
>> You skipped Player turn.
>> Enemy deploys Goblin Barrel
[EFFECT] Goblin Barrel hit Player King Tower dealing 2 damage
[EFFECT] Spawning 2 Goblins on Target!
-----
[ATTACK] Goblin => Player Left Princess Tower (1 dmg)
[COUNTER] Player Left Princess Tower => Goblin (3 dmg)
[DEATH] Enemy Goblin died from counter attack!
[ATTACK] Goblin => Player Right Princess Tower (1 dmg)
[COUNTER] Player Right Princess Tower => Goblin (3 dmg)
[DEATH] Enemy Goblin died from counter attack!
=====

```

Ketika salah satu tower hancur (HP = 0), berikan pesan “[TOWER] <nama_tower> destroyed”.

Enemy

Setelah player memilih salah satu kartu, lawan otomatis akan memilih acak dari antara troop dan. Ketika enemy menggunakan kartu spell prioritas targetnya adalah salah satu dari princess tower lalu king tower (atau semua untuk Arrows). Enemy tidak memiliki elixir.

Game Over

Game akan berakhir jika HP enemy king tower habis, maka player menang. Apabila HP king tower player yang habis, maka player kalah. Setelah game berakhir, kembalikan player ke menu awal.

Cheat

Menginputkan 10 di game akan memaksimalkan elixir player (set ke 10)

Hint

Urutan eksekusi dalam satu turn

- Player memilih aksi (deploy / skip)
- Enemy memilih aksi
- Efek spell langsung dijalankan
- Building aktif milik player memberikan efek
- Building aktif milik enemy memberikan efek
- Troop Player melakukan attack
- Kalau Tower diserang troop, Tower melakukan counter attack
- Cek unit milik player mati & trigger effect
- Troop Enemy melakukan attack
- Kalau Tower Player diserang troop, Tower melakukan counter attack ke Troop Enemy
- Cek unit mati milik enemy & trigger effect
- Cek Game Over
- Tambah elixir player (+1)
- Turn selesai

**DILARANG MENGGUNAKAN KONSEP MAUPUN MATERI YANG BELUM DIAJARKAN
JIKA MELANGGAR MAKA NILAI TUGAS: 0**

PERHATIKAN KETENTUAN DIBAWAH :

- Highlight kriteria yang dikerjakan dengan warna kuning dan kumpulkan word beserta dengan file tugas, apabila tidak dikumpulkan maka tugas tidak akan diperiksa.
- Akan ada pengurangan nilai sebesar -5 untuk setiap kriteria yang dihighlight namun tidak dikerjakan.
- MENCONTEK = Nilai MOD 2

TUGAS : 35

SCORE		KRITERIA
Menu Awal (2)		
0/1		Menampilkan judul game dan pilihan Play & Exit
0/1		Program berhenti saat memilih Exit
Game (9)		
0/1		Player & Enemy memiliki 3 tower dengan HP sesuai
0/1		Tampilan arena (tower + posisi player/enemy) sesuai dengan soal
0/1		Player mulai dengan 5 elixir
0/1		Generate 3 kartu random (Troop, Building, Spell)
0/1		Menampilkan elixir cost tiap kartu
0/1		Player dapat memilih kartu / skip
0/1		Elixir berkurang sesuai cost elixir kartu yang dipilih
0/1		Terdapat validasi elixir (tidak bisa main jika tidak cukup)
0/1		Elixir player bertambah di akhir turn
Kartu Troop (6)		
0/1		Troop spawn sesuai kartu yang dipilih (Goblins, Golem, Bomber)
0/1		Targeting sesuai prioritas (Troop → Building → Tower)
0/1		Troop menyerang setiap turn selama hidup
0/1		Ability Goblins sesuai (bisa spawn banyak)
0/1		Ability Elixir Golem sesuai (bisa pecah jadi 2 saat mati, Mini elixir golem mati bisa memberikan player elixir)
0/1		Ability Bomber sesuai (bisa meledaki semua troop, building, tower lawan saat mati)
Kartu Building (5)		
0/1		Building tetap aktif sampai hancur
0/1		Ability Cannon sesuai
0/1		Ability Elixir Collector sesuai
0/1		Ability Goblin Hut sesuai
0/1		Goblin dari Goblin Hut bisa menyerang enemy
Kartu Spell (6)		
0/1		Spell langsung aktif saat digunakan
0/1		Dapat memilih target untuk Fireball dan Goblin Barrel
0/1		Ability Fireball sesuai
0/1		Ability Arrow sesuai
0/1		Ability Goblin Barrel sesuai
0/1		Goblin dari Goblin Barrel bisa menyerang enemy
Enemy (2)		
0/1		Enemy memilih aksi secara random
0/1		Enemy dapat menggunakan troop / building / spell / skip
Tower (3)		
0/1		Counter attack dari tower berjalan
0/1		HP tower berkurang ketika diserang
0/1		Tower hancur ketika HP = 0
Game Over (1)		
0/1		Game berakhir saat King Tower hancur dan kembali ke menu
Lainnya (1)		
0/1		Cheat berjalan
Total : 35		

Menyetujui

Mengetahui

Penyusun Soal

(Eka Rahayu Setyaningsih,
S.Kom., M.Kom.)
Koordinator Kuliah

(Grace Levina Dewi, M.Kom.)
Koordinator Laboratorium

(Marcella Thamrin)
Asisten